AARON HO
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EDUCATION

University of Southern California **August 2024 to May 2026**
Master of Science in Biomedical Data Analytics | GPA: 3.92/4.00
Relevant Courses: Deep Learning, Neurotechnology, Physiological Signals and Data Analysis, Medical Imaging Systems, Data Analytics, Signals and Systems Analysis

Boston University **August 2018 to May 2022**
Bachelor of Arts with Honors in Biochemistry and Molecular Biology, cum laude

PROFESSIONAL EXPERIENCE

Sanofi, Manufacturing Sciences, Analytics, and Technology (MSAT) **January 2026 to June 2026**
Data Science and Digital Transformation Co-op, Framingham, MA
- Modified and tested a digital twin simulation of Cerezyme perfusion process, modeling 50+ process variables through mass balances, cell growth and substrate-uptake kinetics, gas-liquid mass transfer, and acid-base equations
- Reorganized portions of the modeling code and added automated data inputs, testing, and version-control checks to make simulation runs more consistent and reproducible

Abbott Laboratories, Cardiac Rhythm Management Division (CRM) **May 2025 to December 2025**
Computer Engineering Co-op, Sylmar, CA
- Performed verification and validation testing for an AI-based atrial-fibrillation and pause-detection system used with insertable cardiac monitors, which had successfully analyzed more than 100,000 cardiac episodes within the first two weeks of deployment
- Built a Python and Arduino testing tool with a graphical interface that remotely controlled device orientation across three axes, reducing repeating manual repositioning during validation

Neural Modeling and Brain Interface Lab (Song Lab), USC **January 2025 to May 2025**
Research Assistant, Los Angeles, CA
- Ran Kilosort4 spike sorting on 384-channel Neuropixels recordings sampled at 30 KHz and developed a reproducible workflow for preprocessing, sorting, and unit-level quality control
- Compared 7 Kilsort4 detection-threshold settings using refractory-period violations, firing rates, amplitudes, and signal-quality measurements to identify well-isolated units and investigate questionable clusters

Kwiatkowski Lab, Brigham and Women's Hospital **August 2020 to May 2022**
Research Assistant, Boston, MA
- Investigated two regulators of mTORC1 signaling, DEPTOR and PRAS40, using TSC-deficient cell models, PCR, gel electrophoresis, ELISA, and protein chromatography
- Established in vivo TSC tumor models by engrafting TSC-deficient cells into mice and harvesting tissue for immunofluorescence imaging of DEPTOR and PRAS40 to characterize mTORC1-driven proliferation

PROJECTS

Heart Disease: Multi-Site Clinical ML Study
- Evaluated model transfer across four hospital cohorts (n=920), finding pooled-CV AUC fell from 0.879 to 0.802 under leave-one-hospital-out validation; traced failure modes to label shift, calibration drift, and site-specific missingness
- Built a reproducible clinical-ML audit with nested CV, bootstrap CIs, DeLong tests, decision curves, SHAP, fairness analysis, and conformal abstention; achieved 0.908 coverage at a 90% target with automated tests and CI

GutSense | MIT Grand Hack, 3rd Place (InterSystems Challenge)
- Designed an 8-biomarker colorectal-cancer screening prototype integrating InterSystems IRIS, EHR retrieval, vector RAG, and Claude-generated clinical narratives

Neural Decoding of Imagined Handwriting
- Benchmarked GRU, RCNN, and Conformer-CTC brain-to-text decoders on the Willett handwriting dataset using HMM forced alignment for character labels
- Ran 198 controlled experiments across three seeds and two sequence-length regimes; identity-label errors cost 6.8 points and were 2.4–10.8× more damaging than timing jitter, while architecture choice outweighed label quality about 2:1

TECHNICAL SKILLS
- Programming & tools: Python, SQL, MATLAB/Simulink; PyTorch, scikit-learn, pandas, NumPy
- Methods: Deep learning, time-series analysis, model validation, mechanistic modeling, neural signal processing
- Laboratory: Cell culture, PCR, ELISA, immunofluorescence, protein purification, mouse tumor models

ADDITIONAL EXPERIENCE
- Certified Medical Assistant, Mount Auburn Hospital, Watertown, MA **June 2023 - June 2024**
- AI Data Trainer, Invisible Technologies Inc., Remote **June 2023 - June 2024**